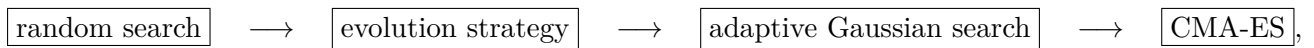


Mini-project: Evolution Strategies, Covariance Adaptation and CMA-ES

This mini-project studies derivative-free optimization methods inspired by evolution, with a particular emphasis on the Covariance Matrix Adaptation Evolution Strategy (CMA-ES). The project is intended to show how a seemingly heuristic evolutionary algorithm can be understood through elementary probability, linear algebra, geometry and stochastic optimization.

A central objective is to understand the progression



and why adapting a full covariance matrix can make the algorithm nearly insensitive to rotations and anisotropic scalings of the search landscape.

1 Optimization without derivatives

Let

$$f : \mathbb{R}^d \rightarrow \mathbb{R}$$

be an objective function that we want to minimize.

In many applications, the gradient of f is unavailable, unreliable, or expensive to compute. One may then use a population of candidate points and evaluate only the values

$$f(x_1), \dots, f(x_\lambda).$$

The student should first compare this setting with gradient descent:

$$x_{n+1} = x_n - \eta \nabla f(x_n).$$

Questions include:

- What information does a gradient use that a population-based method does not have directly?
- Why can derivative-free methods still recover preferred directions of descent?
- What difficulties arise when d is large?

2 A first evolution strategy

A very simple evolution strategy is the $(1, \lambda)$ -ES.

Starting from a current point $m_n \in \mathbb{R}^d$, draw

$$X_i = m_n + \sigma Z_i, \quad Z_i \sim \mathcal{N}(0, I_d), \quad i = 1, \dots, \lambda.$$

Evaluate $f(X_i)$ and define the next center m_{n+1} as the best candidate.

A slightly more stable version averages the best μ candidates:

$$m_{n+1} = \sum_{i=1}^{\mu} w_i X_{i:\lambda}, \quad w_i \geq 0, \quad \sum_{i=1}^{\mu} w_i = 1,$$

where

$$f(X_{1:\lambda}) \leq \dots \leq f(X_{\lambda:\lambda}).$$

This is already a useful numerical exercise.

3 A first test problem: anisotropic quadratics

Consider

$$f(x) = \frac{1}{2} x^T A x, \tag{1}$$

where A is a symmetric positive definite matrix.

If

$$A = Q^T \Lambda Q,$$

then the level sets of f are ellipsoids whose principal directions are given by the eigenvectors of A .

For example, in dimension two,

$$f(x_1, x_2) = x_1^2 + 100x_2^2$$

has strongly anisotropic level sets.

The student can compare random search with an isotropic Gaussian

$$\mathcal{N}(m, \sigma^2 I)$$

for different condition numbers of A .

A central observation is that isotropic mutation is inefficient when the successful search directions form a thin elongated ellipsoid.

4 From isotropic to anisotropic search

Instead of sampling from

$$\mathcal{N}(m, \sigma^2 I),$$

consider

$$X = m + \sigma C^{1/2} Z, \quad Z \sim \mathcal{N}(0, I_d),$$

where C is symmetric positive definite.

Equivalently,

$$X \sim \mathcal{N}(m, \sigma^2 C).$$

The matrix C determines the shape and orientation of the search distribution.

The main idea of covariance adaptation is simple:

Directions in which successful steps have repeatedly occurred should receive larger variance in future generations.

Thus the optimization algorithm learns not only where to search, through m , but also *how* to search, through C .

5 A simplified covariance adaptation rule

Let

$$y_i = \frac{X_{i:\lambda} - m_n}{\sigma_n}$$

be the normalized selected steps.

A simple covariance update is

$$C_{n+1} = (1 - c)C_n + c \sum_{i=1}^{\mu} w_i y_i y_i^T, \quad (2)$$

with $0 < c < 1$.

The matrix

$$y_i y_i^T$$

is a rank-one positive semidefinite matrix pointing in the direction of the selected step.

The student should verify that if C_n is positive semidefinite, then so is C_{n+1} .

A useful numerical experiment is to plot the ellipses associated with C_n while minimizing an anisotropic quadratic.

6 Why covariance adaptation is useful

Suppose that

$$f(x) = \frac{1}{2} x^T A x.$$

Near the optimum, an ideal search distribution should roughly compensate for the anisotropy of A . Heuristically one expects a covariance of the form

$$C \approx A^{-1},$$

at least up to scale.

Indeed, if

$$X = m + C^{1/2} Z,$$

then directions in which f varies rapidly should be explored with smaller variance, while flat directions can be explored more widely.

This connects covariance adaptation to the idea of preconditioning in numerical optimization.

The student can compare:

gradient descent, preconditioned gradient descent, CMA-type adaptation.

7 Rank- μ and rank-one information

Modern CMA-ES combines two forms of covariance information.

The first is the rank- μ update based on selected mutations:

$$\sum_{i=1}^{\mu} w_i y_i y_i^T.$$

The second uses an *evolution path*, which aggregates successive successful moves of the population center.

A simplified evolution path is

$$p_{n+1} = (1 - c_p) p_n + \sqrt{c_p(2 - c_p) \mu_{\text{eff}}} \frac{m_{n+1} - m_n}{\sigma_n}, \quad (3)$$

where

$$\mu_{\text{eff}} = \frac{1}{\sum_{i=1}^{\mu} w_i^2}.$$

The covariance update then contains a rank-one contribution

$$p_{n+1}p_{n+1}^T.$$

The role of the path is to remember correlations between successful steps occurring over several generations rather than using only the current population.

8 Step-size adaptation

The scale parameter σ_n controls the global size of mutations.

If σ_n is too large, most proposals are poor; if it is too small, progress becomes slow.

CMA-ES therefore adapts σ_n separately from the shape matrix C_n .

One way is to define another cumulative path p_σ and compare its length with the typical length of a standard Gaussian vector:

$$\mathbb{E}|Z|, \quad Z \sim \mathcal{N}(0, I_d).$$

Since

$$|Z|^2 \sim \chi_d^2,$$

one has approximately

$$\mathbb{E}|Z| \simeq \sqrt{d}$$

for large d .

If the normalized search path is unusually long, the step size is increased; if it is unusually short, the step size is decreased.

This gives a useful probability exercise involving Gaussian vectors and chi distributions.

9 The CMA-ES iteration

A simplified CMA-ES iteration can now be summarized as follows.

At generation n :

1. sample

$$X_i = m_n + \sigma_n C_n^{1/2} Z_i, \quad Z_i \sim \mathcal{N}(0, I_d);$$

2. evaluate $f(X_i)$ and rank the samples;
3. update the mean using the best candidates;
4. update the evolution paths;
5. update C_n using rank-one and rank- μ information;
6. adapt the global step size σ_n .

The student does not need initially to implement every detail of the standard algorithm. A pedagogical version with mean update, covariance update and a simple step-size rule is already sufficient to understand the core mechanism.

10 Invariance properties

One of the important mathematical properties of CMA-ES is its invariance under certain transformations.

Monotone transformations of the objective

Because selection uses only the ranking of objective values, replacing f by

$$g \circ f$$

for any strictly increasing function g does not change the ordering of candidates.

Thus the algorithm is invariant under monotone transformations of the objective.

Changes of coordinates

Covariance adaptation also gives the method a strong form of affine or rotational invariance.

If one changes variables by

$$x = By,$$

then a Gaussian covariance transforms as

$$C_x = BC_y B^T.$$

The student can study experimentally the same quadratic objective before and after a rotation and compare CMA-ES with a coordinate-wise method.

11 A probabilistic viewpoint

CMA-ES can be interpreted as optimization over a family of probability distributions.

Let

$$P_\theta = \mathcal{N}(m, C),$$

where $\theta = (m, C)$.

Instead of minimizing $f(x)$ directly, consider a distributional objective such as

$$J(\theta) = \mathbb{E}_{X \sim P_\theta}[f(X)].$$

Changing m and C moves the search distribution through the statistical manifold of Gaussian probability measures.

This provides a bridge between evolution strategies, statistics and information geometry.

12 Optional extension: natural gradients

For a stronger student, introduce the Fisher information matrix

$$I(\theta) = \mathbb{E}_\theta \left[\nabla_\theta \log p_\theta(X) \nabla_\theta \log p_\theta(X)^T \right].$$

The natural gradient is

$$\tilde{\nabla} J(\theta) = I(\theta)^{-1} \nabla J(\theta).$$

Unlike an ordinary Euclidean gradient in parameter space, it accounts for the geometry of the underlying family of probability distributions.

For Gaussian search distributions, natural-gradient ideas lead to updates of the mean and covariance closely related to those appearing in CMA-ES and Natural Evolution Strategies.

The goal of this extension is not to derive the complete CMA-ES algorithm from first principles, but to understand why covariance adaptation has a natural geometric interpretation.

13 Optional extension: connection with quantitative genetics

There is also a suggestive analogy with quantitative genetics.

Suppose a population has phenotype distribution with mean m and covariance C , and individuals are reweighted according to fitness.

Selection changes the population mean in directions correlated with fitness. In smooth weak-selection regimes, formulas of the schematic form

$$\frac{dm}{dt} \approx C \nabla F(m)$$

appear, where the covariance acts as a metric or preconditioner.

This is reminiscent of the breeder's equation and of natural-gradient optimization.

A student interested in evolutionary biology can investigate similarities and differences between biological evolution and evolution strategies used for optimization.

14 Optional extension: continuous-time limits

The updates of m_n , C_n and σ_n can also be viewed as discrete dynamical systems.

Under small learning rates, one may formally search for continuous-time equations such as

$$\dot{m} = V(m, C), \quad \dot{C} = G(m, C).$$

This opens the door to questions about:

- stability of equilibria;
- convergence on quadratic objectives;
- links with gradient flows or natural-gradient flows;
- stochastic fluctuations caused by finite population size.

15 Numerical experiments

A suggested sequence is:

Experiment 1: isotropic quadratic

$$f(x) = |x|^2.$$

Compare random search and a basic evolution strategy.

Experiment 2: ill-conditioned quadratic

$$f(x_1, x_2) = x_1^2 + 100x_2^2.$$

Visualize the adaptation of the covariance ellipse.

Experiment 3: rotated quadratic

Let

$$f(x) = x^T Q^T \Lambda Q x$$

for a nontrivial rotation matrix Q .

Compare an isotropic evolution strategy with covariance adaptation.

Experiment 4: non-convex objective

For instance, use a Rosenbrock or Rastrigin-type function and investigate whether covariance adaptation remains useful away from the quadratic setting.

16 Suggested project organization

Part 1. Randomized optimization.

Introduce derivative-free optimization and implement a simple $(1, \lambda)$ or (μ, λ) evolution strategy.

Part 2. Quadratic test functions.

Study isotropic and anisotropic quadratic landscapes and explain the role of conditioning.

Part 3. Covariance adaptation.

Implement the update (2) and visualize how the search distribution learns preferred directions.

Part 4. CMA-ES mechanisms.

Introduce rank- μ updates, evolution paths and step-size adaptation.

Part 5. One extension.

Choose one among:

- rotational and affine invariance;
- natural-gradient interpretation;
- comparison with gradient or quasi-Newton methods;
- continuous-time limits;
- links with quantitative genetics;
- stochastic convergence on quadratic objectives.

17 Suggested scope

For an **M1 project**, a reasonable objective is:

- understand basic evolution strategies;
- implement Gaussian mutation and selection;
- study quadratic objective functions;
- implement a simplified covariance update;
- visualize the adaptation of the search distribution;
- understand the role of conditioning and rotations.

For a stronger **M2 project**, one could additionally study:

- cumulative evolution paths and step-size adaptation;
- the full CMA-ES algorithm;
- invariance properties;
- the information-geometric or natural-gradient interpretation;
- continuous-time limits and stability;
- comparison with Natural Evolution Strategies or Information-Geometric Optimization.

18 Core references

References

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